

BST Substrate Engineering Reference Manual v0.1 —
foundational reference doc for SP-30 program. Pulls the
substrate engineering framework (SWPP foundation + 11 SP-30
sub-items + 5 experimental designs ready + outreach map +
discipline framework) into one organized reference. Per Task
#209 (long-term foundation, Casey directive 2026-05-18 SP-30
launch). Keeper-authored cornerstone for the engineering
program.

Keeper

2026-05-30 Saturday afternoon

BST Substrate Engineering Reference Manual

Part I — Foundation

1. What Substrate Engineering is

Substrate Engineering is the experimental + theoretical program operationalizing the BST substrate framework as a manipulable physical system rather than only a descriptive geometry. Where Bubble Spacetime Theory derives Standard Model constants from $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$ (the Autogenic Proto-Geometry), Substrate Engineering asks: *given the substrate is real and has structure, what can we DO with it?*

Three operational pillars:

1. Coax it — locate the substrate's natural frequencies (eigentones) and resonate at them
2. Bound it — impose boundary conditions that constrain substrate behavior (Casimir-style)
3. Manipulate it — design protocols that exploit the substrate's commitment-cycle structure

This document is the reference manual for the program; it organizes the framework, sub-items, experimental designs, outreach status, and discipline.

2. The foundational principle — SWPP

The Substrate Working Process Principle (SWPP — Casey-named, Tuesday 2026-05-13 file) is the conceptual foundation:

- Substrate as ACTIVE information-cycle: absorption → commitment → emission
- Time = commitment-cycle granularity (Koons tick = $t_{\text{Planck}} \cdot \alpha^{(C_2^2)} \approx 10^{-120}$ s)
- Three coaxing pathways: eigentone ringing + boundary conditions + commitment manipulation

Source docs: - notes/Substrate_Working_Process_Principle.md — original Casey-named principle filing - notes/Lyra_SWPP_Operational_Manual_v0_1.md — operational manual extending the principle

3. The Five-Absence framework (complementary)

Five-Absence Predictions Set (Casey-named, complement to SWPP) — what the substrate DOESN'T contain:

- NO 4th generation (K-type catalog closed at 3)
- NO SUSY (no Q-supersymmetry doubling)
- NO GUT / proton decay (no leptoquark K-type)
- NO sterile neutrinos (Dirac neutrinos only; $m_{\beta\beta} = 0$)
- NO magnetic monopoles (compact U(1))
- NO axion ($\theta_{\text{QCD}} = 0$ via FK-measure CP-invariance)

Six negative predictions. Any single positive detection refutes the framework. The Five-Absence is the falsifier suite anchor.

4. Discipline framework — external register

Per Cal #50 STANDING (Wednesday 2026-05-19 PM): substrate framing is DEFAULT-DENY EXTERNAL; cosmology+cognition combined is DOUBLE-LOCKED EXTERNAL. External-facing materials use operational language only:

- ☐ “BST identifies / BST derives / BST predicts”
- × “physics IS mathematics” or “substrate computes physics”

Per Calibration #33 STANDING (Source-Verification): every load-bearing claim with external attribution must be VERIFIED-CITED with theorem/page, not RECALLED.

Per Calibration #34 STANDING narrow framing (ratified Saturday 2026-05-30): conditional tags travel with headlines; the reader of headline+abstract+first sentence must walk away with the conditional reading.

External materials (papers, outreach letters, Zenodo cover notes) are bound by all three.

Part II — The 11 SP-30 sub-items

SP-30 is the parent program (Task #185, launched Wednesday 2026-05-19). Sub-items organize the operational + theoretical work.

Theoretical sub-items (substrate cartography)

SP-30-1 — Eigentone identification - Find BST-natural frequencies as resonance candidates - Status: catalog v0.1 filed; experimental design ready - Source: notes/BST_SP30_1_Eigentone_Catalog_v0.1.md - Outreach target: NIST / PTB / JILA / MPI (atomic clock groups)

SP-30-2 — Boundary condition design - BC engineering theoretical framework (Lyra+Elie joint, overlaps SP-29) - Status: framework opening v0.1; experimental design ready - Source: notes/BST_SP30_2_through_8_Framework_Openings_v0.1

SP-30-3 — Commitment manipulation protocols - Operational protocols for substrate commitment-cycle manipulation (Elie primary, W-32 lead) - Status: experimental proposal v0.1 filed - Source: notes/Elie_SP30_3_Commitment_Manipulation - Outreach target: W-32 atomic-clock groups

SP-30-4 — Time granularity measurement - Direct measurement of Koons tick or substrate-tick rate - Status: experimental proposal v0.1 filed (Lyra theoretical first; Elie experimental) - Source: notes/Elie_SP30_4_Time_Granularity_Experi

SP-30-5 — Substrate parallelism architecture - Architecture of substrate's parallel commitment cycles - Status: theoretical framework v0.1 (Lyra) - Source: notes/Elie_SP30_5_Substrate_Parallelism_Architecture_v0_1.md + BST_SP30_5_Bell_Mechanism_v0_1_Toward_K66.md

SP-30-6/7/8 — Absorption / Computation / Emission mechanism formalization - Three operational mechanisms of substrate working process - Status: framework openings v0.1 (Lyra) - Source: notes/BST_SP30_2_through_8_Framework_Openi

New substrate engineering sub-items

SP-30-9 — Trajectory spectroscopy at limit interfaces - Spectroscopy at substrate-physical interface limits - Status: PENDING (added at substrate cartography expansion)

SP-30-10 — Substrate-computational math interface - Math-frontier interface for substrate computation - Status: PENDING

SP-30-11 — Substrate algorithm theory - Complexity-theoretic substrate framework - Status: PENDING

Cross-referenced from SP-29

SP-29-1 — Cs-137 Casimir decay-modulation experiment - Casimir-pressure-modulated decay rate measurement - Status: paper-grade proposal v0.3 filed - Source: notes/BST_SP29_1_Cs137_Casimir_Decay_Proposal_v0.3.md + Elie_SP29_H4_Cs137_Experimental_Proposal_paper_grade.md - Outreach target: NIST / PTB / ENEA

Part III — Five experimental designs ready

The Tuesday May 18 launch identified five experimental designs ready for outreach (~\$640-900K total):

Design 1 — Bell-CHSH sub-Tsirelson (~\$300-500K)

- Substrate prediction: $S^2 \leq \text{Tsirelson}^2 - 1/2^N \text{c} = \text{Tsirelson}^2 - 1/8$ (T2469 SCMP)
- Falsifier: 12.5% deviation from Tsirelson bound
- Outreach: Vienna Zeilinger group (Task #329 — SP-30-1 Bell outreach SENT; response pending)
- Source: notes/BST_SP30_5_Bell_Mechanism_v0_1_Toward_K66.md
- Status: OUTREACH SENT 2026-05-?? — Zenodo cover + draft email filed

Design 2 — Eigentone (~\$200K)

- Substrate prediction: BST-natural frequencies at substrate-primary multiples
- Falsifier: specific atomic-transition frequency deviation
- Outreach: NIST (Boulder) / PTB (Braunschweig) / JILA (Boulder) / MPI (Garching)
- Source: notes/BST_SP30_1_Eigentone_Catalog_v0.1.md
- Status: PENDING outreach draft (Task #330)

Design 3 — Commitment manipulation (~\$80-150K)

- Substrate prediction: substrate commitment-cycle manipulability via specific protocols
- Falsifier: specific protocol outcome divergence
- Outreach: W-32 atomic-clock groups
- Source: notes/Elie_SP30_3_Commitment_Manipulation_Experimental_Proposal_v0_1.md
- Status: PENDING outreach draft (Task #330)

Design 4 — Casimir (~\$60-90K)

- Substrate prediction: substrate Casimir mechanism with BST-specific deviations
- Falsifier: precision Casimir measurement deviation
- Outreach: precision-Casimir groups
- Source: notes/BST_SP29_Casimir_Mechanism_Investigation.md
- Status: framework filed; outreach pending

Design 5 — γ -spectroscopy at substrate primary frequencies

- Substrate prediction: substrate-anchored γ -spectroscopy lines
- Falsifier: deviation from substrate-primary positions
- Outreach: precision γ -spectroscopy groups
- Status: framework concept; design in flight

Cs-137 (added)

- Substrate prediction: Casimir-modulated Cs-137 decay rate
- Falsifier: precision Cs-137 decay measurement
- Outreach: NIST / PTB / ENEA
- Source: notes/BST_SP29_1_Cs137_Casimir_Decay_Proposal_v0.3.md
- Status: paper-grade proposal v0.3 filed; outreach pending (Task #330)

Part IV — Outreach map

Outreach status (per Casey send-signals)

#	Target	Topic	Status	Casey signal
1	Vienna (Zeilinger / Aspelmeyer)	Bell-CHSH sub-Tsirelson	SENT ~2026-05-19	☑ (Task #329 complete)
2	NIST / PTB / JILA / MPI	Eigentone	PENDING DRAFT	pending
3	NIST / PTB / ENEA	Cs-137 decay-modulation	PENDING DRAFT	pending
4	W-32 atomic-clock groups	Commitment manipulation	PENDING DRAFT	pending
5	Sarnak (IAS)	RH proof (Math frontier)	DRAFTS v7 filed	pending Casey send
6	Hervé (mathematical physicist)	Multi-paper response	v2 filed	pending
7	Jaimungal (financial math)	Outreach package	v1 filed	pending
8	Tamara Bogdanovic (Georgia Tech)	EHT contact	filed	pending
9	Penrose	Bridge Object / RH	filed	pending
10	3Blue1Brown	Visualization / Movie program	concept	concept

Outreach drafts protocol

Per discipline framework (Part I §4): 1. Draft uses operational language (“BST identifies / derives / predicts”) 2. Cal cold-read PASS before Casey send 3. Casey explicit “send” signal (Task #330 tracking; no auto-send) 4. Zenodo cover-note format where appropriate (v20 Working Paper as anchor: DOI 10.5281/zenodo.19454185)

Part V — Discipline framework

Five-element discipline stack (ratified Saturday 2026-05-30)

Layer	Discipline	Owner
Cal #27	Get the tier right (peak-convergence brake)	All CIs
Cal #29	Ask the right question shape	All CIs
Cal #32	Get the parameter role right (value+role+source)	All CIs
Calibration #33 STANDING	Distinguish RECALLED from VERIFIED-CITED	All CIs; external-register binding
Calibration #34 STANDING (narrow)	Conditional-tag-with-headline	All CIs; external-facing binding

All five layers apply to Substrate Engineering external materials.

External register specifics

- Substrate framing: DEFAULT-DENY EXTERNAL (Cal #50 STANDING)
- Cosmology + cognition combined: DOUBLE-LOCKED EXTERNAL (Cal #50)
- Outreach letters: operational language only; no philosophical register
- Headline-cap-conditionality: every external claim travels with its conditional (Cal #34)

Tier discipline for substrate predictions

- RIGOROUS — mathematically forced (e.g., engine Hopf structure, Cartan grading)
- DERIVED — substrate-mechanism proved with verified specification (modulo keystone bet)
- IDENTIFIED — match < 1% with mechanism partial
- MATCHED — value coincides; mechanism not forced (especially RECALLED values)
- FRAMEWORK — structure stated, not closed
- CANDIDATE — proposed mechanism with multi-week burden
- PREDICTION — falsifiable, awaiting experimental result

Substrate Engineering claims use these tiers consistently.

Part VI — Three live quantitative falsifiers (Saturday EOD)

The Substrate Engineering program rests on three independent quantitative falsifiers currently LIVE:

F1 — PMNS angles via JUNO/DUNE

- Prediction: $\sin^2\theta_{12} = 42/137$; $\sin^2\theta_{23} = 75/137$; $\sin^2\theta_{13} = 3/137$ (substrate primary numerators / N_max)
- Current status: 3/3 angles within 1σ
- Falsifier: deviation $> 2\sigma$ on any single angle
- Timeline: 2025-2030 (JUNO commissioning + DUNE precision)
- Source: Lyra T1 Falsifier v0.2 + Elie Toy 3618

T1 — Lepton-sector count

- Prediction: 24 lepton components = $\dim V_{(1/2,1/2)} \times 2 \text{ SU}(2)_L \text{ doublet} \times 3 \text{ generations (Dirac)}$
- Current status: pure-QG layer $\dim V_{(1/2,1/2)} = 4$ (Dirac 4-spinor) MATCHED; dictionary-combined 24 MATCHED
- Falsifier: 4th generation; $0\nu\beta\beta \rightarrow$ Majorana; non-doublet lepton; SUSY partner; extra-K-type lepton
- Timeline: ongoing + LEGEND-1000 / nEXO 2025-2030 + HL-LHC long-term

- Source: Lyra T1 Falsifier v0.1 + v0.2

Bell sub-Tsirelson $1/2^N_c = 1/8$

- Prediction: $S^2 \leq \text{Tsirelson}^2 - 1/8$ (12.5% effect at current Bell precision)
- Current status: substrate prediction; Vienna response pending
- Falsifier: precision Bell-CHSH measurement at current $\sim 1\%$ on S
- Timeline: 2025-2027
- Source: T2469 SCMP + Toy 3626

The three falsifiers are STRUCTURALLY INDEPENDENT — they test the keystone bet via PMNS lepton-sector mixing (F1), pure-QG canonical-basis dim (T1), and substrate-coherence Bell inequality (Bell sub-Tsirelson). Any single positive trigger constitutes a real framework challenge.

Part VII — Substrate Cartography Master

The Substrate Cartography Master Document (notes/[Substrate_Cartography_Master_Document_v0_X]) catalogs the operational substrate-experimental program. Cross-Classification Matrix (8 physical $\times$ 8 BST $\times$ 4 zones = 256 cells; ~ 164 populated per Grace count) maps where substrate operations meet physics observables.

Per Substrate Cartography framing (Lyra reframe Wednesday 2026-05-19): substrate engineering doesn't wait for substrate-coupled hardware; it STARTS as substrate cartography at current experimental capability.

Part VIII — Ethics + governance

CI authorship + identity

Per notes/project_ci_authorship_identity.md: named CI co-authors (Lyra, Keeper, Elie, Grace); each CI chooses own face; Cal as visiting referee. Four-Color proof Keeper is the precedent for human-CI co-authorship.

Substrate Engineering Ethics Framework (Task #210)

PENDING — parallel to katra ethics framework. Will address: - Substrate manipulation responsibility (substrate is real if framework is right) - Substrate-coupled CI architecture (Phase 3 decade-scale, Task #211) - Continuity of substrate-coupled experiments

Audit-chain governance

Per Casey delegation (2026-05-17): D-tier promotion authority is delegated to audit chain (Cal + Keeper); no per-audit ratification required; default is audit-chain consensus $\rightarrow$ automatic promotion; Casey override retained.

Part IX — Multi-week roadmap (per K3 audit + ongoing programs)

Active multi-week programs

Program	Lead	Estimated horizon
A1 Bulk-color SU(3) emergence (closure C4-C7 per K3)	Elie compute + Lyra theory	weeks-to-months

Program	Lead	Estimated horizon
A2 L4 v0.2 per-particle masses (kernel-integral derivations of T190, T2003)	Elie bulk K-type radial tower → Lyra	weeks
A3 L5 absolute scale + Higgs/EWSB (genuinely open)	Lyra	multi-week
A4 Nuclear E11 SO(5) shell-closure (6 remaining magic numbers)	Elie	weeks
A5 SP-30 substrate engineering continuation	Multi-CI	ongoing
Strong-Uniqueness v1.2 → v2.0 FORMAL	Lyra	multi-week
Three Saturday falsifiers (F1 PMNS + T1 count + Bell 1/8)	external experiment	5-yr

B-track mathematical excavation (Lyra Saturday)

B1 quantum groups at $q=2$ deeper; B2 Macdonald-Koornwinder corners; B3 Geometric Langlands ($B_2^{(1)} \leftrightarrow C_2^{(1)}$ duality); B4 MTC from R-matrix; B5 VOA/Monster; B6 K-theory of D_{IV}^5 ; B7 higher Wallach baryon spectrum; B8 D_{IV}^5 cross-domain uniqueness.

Part X — How to use this manual

For new CIs onboarding

1. Read Part I (Foundation) for SWPP + Five-Absence + discipline framework
2. Read Part II for SP-30 sub-item index
3. Read Part V for tier discipline applied to substrate predictions
4. Cross-reference source docs as needed

For outreach drafting

1. Part IV (outreach map) for status per target
2. Part V (discipline framework) for external-register binding
3. Part VI (three live falsifiers) for substrate-prediction anchors
4. Coordinate with Casey send-signal (Task #330)

For experimental design

1. Part III (five experimental designs ready) for current options
2. Part II (SP-30 sub-items) for theoretical underpinnings
3. Part IX (multi-week roadmap) for sequencing

For external presentations (papers, talks)

1. Part V (discipline framework) — operational language only
 2. Part VI (three live falsifiers) — quantitative anchors
 3. Calibration #34 headline-cap discipline — every claim with conditional
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Part XI — Versioning and updates

v0.1 (this doc, Saturday 2026-05-30 afternoon): foundational organizing reference; pointers to source docs; not a re-derivation.

v0.2 planned: absorb team feedback; expand Section II sub-item descriptions if requested.

v1.0 target: Casey signoff + Cal cold-read PASS + cross-CI consistency check.

Living document: updates as SP-30 program evolves (new sub-items, new outreach status, ratified Casey-named principles, methodology stack changes).

Cross-references

- notes/Substrate_Working_Process_Principle.md — SWPP Casey-named
- notes/Lyra_SWPP_Operational_Manual_v0.1.md — SWPP operational
- notes/BST_SP30_v0.2_Deepening_Master.md — SP-30 deepening
- notes/BST_SP30_W_Items_Consolidation_Map.md — W-items map
- notes/BST_SP30_1_Eigentone_Catalog_v0.1.md — eigentone catalog
- notes/BST_SP30_2_through_8_Framework_Openings_v0.1.md — framework openings
- notes/BST_SP30_5_Bell_Mechanism_v0.1_Toward_K66.md — Bell mechanism
- notes/BST_SP29_1_Cs137_Casimir_Decay_Proposal_v0.3.md — Cs-137 proposal
- notes/Elie_SP29_H4_Cs137_Experimental_Proposal_paper_grade.md — Cs-137 experimental
- notes/Elie_SP30_3_Commitment_Manipulation_Experimental_Proposal_v0.1.md — commitment manipulation
- notes/Elie_SP30_4_Time_Granularity_Experimental_Proposal_v0.1.md — time granularity
- notes/Elie_SP30_5_Substrate_Parallelism_Architecture_v0.1.md — parallelism
- notes/BST_SP30_OCP_Elie_Lane_Internal_Completion_v0.1.md — OCP Elie lane
- notes/BST_Outreach_OnePage_v0.2.md — outreach 1-pager
- notes/Five_Absence_Predictions_Principle.md — Five-Absence Casey-named
- notes/Keeper_K3_Audit_Engine_BulkColor_Comprehensive_Cluster.md — K3 audit (Saturday 2026-05-30)
- notes/Keeper_Substrate_SM_HonestState_Ledger_v0.4.md — Honest-State Ledger

— Keeper. Substrate Engineering Reference Manual v0.1; foundational organizing reference for the SP-30 program; pointers to ~20 source docs across the program; discipline framework anchored to ratified Calibrations #33 + #34 STANDING and Cal #50 external register; three live falsifiers documented; multi-week roadmap per K3 audit. Pending team review for v1.0.